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Research paper

Intelligent decision and planning for unmanned surface vehicle: A review of machine learning techniques

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Highlights

- Traced evolutionary milestones of machine learning (ML) in USV autonomy research.
- A ML-MDP is proposed, offering unified theoretical guidance for USV intelligent decision and planning.
- Provided a overview of recent research progress in ML for USV mission planning, dynamic decision, and path planning.
- Analyzed the challenges and opportunities of ML applications in USV decision and planning.

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Abstract

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Abstract

With the increasing demand for unmanned surface vehicles (USVs) in fields such as marine environmental monitoring, resource exploration, and emergency rescue, the development of intelligent decision and planning technologies has become critical. However, the complexity and dynamic nature of marine environments pose significant challenges to traditional methods in practical applications. In recent years, the rapid advancement of machine learning (ML) has offered novel solutions for the intelligent decision and planning of USVs. This paper systematically reviews the research progress in USV decision and planning based on ML. First, it reviews the classification of USV autonomy levels and the historical development of ML in unmanned marine systems. Then, the paper proposes and elaborates on the "ML-MDP" framework (a ML-based Mission planning, Dynamic decision, and Path planning framework) for USVs, analyzing the latest research outcomes in these areas and explores the suitability of various ML algorithms in addressing these challenges. Finally, the paper analyzes the challenges faced by ML in USV applications and its future development directions. This review aims to provide a valuable reference for researchers in related fields, highlighting the potential of ML in marine unmanned systems and promoting advancements in USV intelligence.

Keywords

USV; Machine learning; Mission planning; Dynamic decision; Path planning

Abbreviations

AC, actor-critic; AI, artificial intelligence; AIS, automatic identification system; ANN, artificial neural network; APF, artifical potential field; ARM, auto regression model; B&B, branch-and-bound algorithm; BN, Bayesian network; BP, back propagation neural network; CA, clustering algorithm; CFD, computational fluid dynamics; CNN, convolutional neural network; CNP, contract net protocol; COLREGs, international regulations for preventing collisions at sea; CTDE, centralized training and decentralized execution; DBSCAN, density-based spatial clustering of applications with noise; DDPG, deep deterministic policy gradient; DDQN, double deep Q-network; DL, deep learning; DPG, deterministic policy gradient; DQN, deep Q-network; DRL, deep reinforcement learning; FCNN, fully connected neural network; FL, federated learning; FMM, fast marching method; GA, genetic algorithms; GAE, generalized advantage estimation; GAN, generative adversarial network; GMM, Gaussian mixture model clustering; GNN, graph neural networks; GPR, Gaussian process regression; GRU, gated recurrent unit network; GWO, grey wolf optimization; HA, hungarian algorithm; IMO, international maritime organization; KF, Kalman filter; LLM, large language model; LSTM, long short-term memory network; MASS, maritime autonomous surface ship; ML, machine learning; ML-MDP, ML-based mission planning, dynamic decision, and path planning framework; MP, motion prediction; MPC, model predictive control; MSC, maritime safety committee; M-STCP, matrix neural network-based online ship track cleaning and prediction algorithm; PG, policy gradient; PPO, proximal policy optimization algorithm; PSO, particle swarm optimization; RF, random forest; RL, reinforcement learning; RNN, recurrent neural network; RRT, rapid random tree; SAC, soft actor-critic; SAM, self-attention mechanism; SARSA, state-action-reward-state-action algorithm; Seq2Seq, sequence-to-sequence; SOM, self-organising maps algorithm; STGCN, spatio temporal graph convolutional network; SVM, support vector machine; SVR, support vector regression; TD, temporal difference; TD3, twin delayed deep deterministic policy gradient; TSP, traveling salesman problem; USV, unmanned surface vehicle; VG, visibility graph; VO, velocity obstacle; WOA, whale optimization algorithm

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Introduction

Covering approximately 70 % of Earth's surface, the ocean holds abundant resources and immense development potential. With the rapid advancement of marine science and technology, unmanned surface vehicles (USVs) are playing an increasingly critical role in marine environmental monitoring, resource exploration, and maritime security. USVs possess the capability to operate autonomously without human intervention. Their intelligent decision and planning refer to the process of applying intelligent algorithms to analyze mission requirements during autonomous navigation and operations, dynamically selecting the optimal action strategy based on environmental perception, and generating executable paths under multiple constraints. This process encompasses three core components: mission planning, dynamic decision, and path planning. Its significance lies in enhancing the safety and adaptability of USVs in complex maritime environments. However, the highly dynamic, complex, and uncertain nature of the

marine environment poses significant challenges to their autonomous navigation and operations (Wu et al., 2022). Intelligent decision-making and planning capabilities are critical for USVs to achieve embodied intelligence and successful deployment in complex and dynamic marine environments (Ivanov, 2023). These capabilities enable USVs to autonomously navigate, avoid obstacles, and perform complex tasks in such environments, significantly enhancing their autonomous navigation capacity, as well as the efficiency and success rate of mission execution. Traditional methods, including rule-based systems, pre-set instructions, heuristic algorithms, game theory, optimization algorithms, and statistical analyses, have been widely applied in the decision-making systems of USVs (Tang et al., 2022). However, these methods often struggle to address the high dynamics, complexity, and uncertainty of marine environments. To enhance the autonomy and intelligence of USVs, researchers are continuously exploring new methods and technologies, aiming to develop more robust and efficient intelligent decision-making and planning systems to meet the practical demands of marine environments.

In recent years, advances in computational power and big data have demonstrated the superiority of machine learning (ML) (Santiago and Ortiz-Rivera, 2023), including deep learning (DL) (Ahn and Lee, 2022) and reinforcement learning (RL) (Clemmons and Jin, 2023), in various domains such as image recognition (Zhou et al., 2023a), natural language processing (Lauriola et al., 2022), and autonomous driving (Grigorescu et al., 2020). Due to their robust adaptability and generalization capabilities, ML algorithms have been increasingly applied in marine science and engineering. USV intelligent decision-making and planning in complex marine environments are characterized by high-level, strong non-linearity and high coupling (Ardon et al., 2024). ML offers several advantages over traditional algorithms. It excels in processing dynamic marine data and providing real-time decision support. Additionally, it is highly effective in recognizing complex patterns, optimizing multiple objectives, and predicting intelligent behavior. Through continuous interaction with and feedback from the environment, ML enables USVs to adaptively learn optimal decision-making strategies in highly uncertain and dynamic conditions, improving both autonomy and responsiveness.

Several scholars have reviewed the applications of USVs in various fields. Ghazali et al. (2024) reviewed research and practices on USV hull design from the perspective of hydrodynamic performance. Xing et al. (2023b) summarized recent developments in foundational algorithms for USV path planning. Zhou et al. (2020) discussed the progress of multimodal constraints in USVs and provided a detailed analysis of kinematic and dynamic constraints. Despite significant advancements in USV intelligence and the success of ML applications, current research primarily focuses on fluid analysis (Wang et al., 2016), digital twin (Madusanka et al., 2023), basic planning (Wang et al., 2022b), and tracking control (Yi et al., 2020). As USVs evolve from nearshore to open-ocean operations, and missions become more complex, moving from single-vehicle navigation to fleet coordination (Wang et al., 2023), more intelligent decision and planning will become increasingly critical and urgent.

Overall, research on USV decision-making and planning based on ML has made significant progress over the past decade, providing crucial technical support for the development of USV intelligence and autonomy. This paper focuses on ML-driven intelligent decision and planning for USVs, incorporating decision-making and planning into a unified framework. By systematically reviewing recent research advancements in the field, it highlights the applications of ML in USV dynamic decision, mission planning, and path planning, as well as the challenges faced. Additionally, it provides an outlook on future research directions and development trends. By reviewing recent

research advances, this paper aims to provide researchers and engineers in the field with a comprehensive perspective, promoting the integration and innovation of interdisciplinary knowledge and advancing USV technology toward greater intelligence and autonomy. Compared to existing studies, the main contributions of this paper are as follows:

(1) The study reviews the trajectory of ML applications in USVs, uncovering key pathways in their evolution toward intelligence.

(2) A ML based-Mission planning, Dynamic decision, and Path planning framework (ML-MDP) is proposed, offering unified theoretical guidance for USV intelligent decision and planning.

(3) The paper comprehensively reviews recent ML applications in mission planning, dynamic decision, and path planning, evaluating the strengths, limitations, and applicability of various algorithms to guide USV intelligence.

(4) Key challenges in USV decision and planning, such as data scarcity, computational complexity, and integrating physical constraints are analyzed, while emerging technologies like federated learning, knowledge distillation, and digital twins are proposed as pathways toward clustered and collaborative intelligent ocean systems.

The remainder of the paper is structured as follows. Section 2 introduces the autonomy levels of USVs and the development trajectory of ML in the field of marine unmanned equipment. Section 3 presents a ML-based intelligent decision and planning framework for USVs. Sections 4–6 elaborate on the latest research advancements in ML for USV mission planning, dynamic decision, and path planning, respectively. Section 7 addresses future research opportunities and challenges. Section 8 concludes the paper.

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Autonomy class

A USV refers to various types of unmanned surface vehicles (Li et al., 2023b), equipped with a range of sensors, control technologies, and automation systems to execute a

variety of maritime tasks, such as environmental monitoring, resource exploration, and scientific research. Common used of USVs as shown in Fig. 1.

The autonomy level of a USV is a critical metric (Han et al., 2022) for evaluating its ability to autonomously make decisions, perceive, plan, and control across various tasks and …

Basic framework

The decision and planning problem of USVs is essentially central to achieving autonomous behavior (Liu et al., 2024c). With intelligent decision and planning, USVs can make optimal choices based on environmental information and task requirements and follow appropriate action paths (Lyu et al., 2022). Planning provides a constraint space for decision-making, while decision-making injects dynamic adaptability into planning. Building an effective foundational framework is a crucial step in …

USV mission planning

Mission planning establishes a global mission framework for USVs, forming the goal-oriented foundation of the decision and planning system. Mission planning is a crucial module for intelligent decision-making in USVs. For each USV, it involves the allocation of complex tasks by considering factors such as task type, distance, and priority. In fact, mission planning is a prerequisite for enabling USVs to carry out complex marine operations. When dealing with scenarios involving multiple …

USV dynamic decision

Dynamic decision enables the USV system to respond in real-time to environmental changes, reflecting the core intelligent value of decision-making and planning in uncertain scenarios. USV dynamic decision is a real-time decision-generation mechanism designed for dynamic marine environments, with its core logic being a progressive dynamic decision process that connects behavior prediction, risk assessment, and action decision-making (LaValle, 2006). Behavior prediction integrates other vessels' …

USV path planning

Path planning transforms higher-level decisions into spatial motion trajectories, forming a key technological link between decision logic and physical execution. The core task of path planning is to find the optimal route for a USV from the starting point to the target point (Zhou et al., 2022a), ensuring that it avoids obstacles while meeting various constraints (Zhao et al., 2024c) such as energy consumption (Wu et al., 2023c), time, safety, and regulations. Considering these constraints …

Challenges

In recent years, with the development of AI (Artificial Intelligence), ML has gradually been introduced into marine unmanned systems, driving the rapid advancement of USV intelligence. This has led to significant progress in decision-making, perception, planning, and control. However, USVs still generally operate at a relatively low level of autonomy, and their path to widespread commercialization faces several challenges:

(1) Real-Time Performance and Computational Resource Limitations

When …

…

Conclusion

Over the past decade, USVs have made significant progress in their autonomy and intelligence, and ML-related technologies have also seen a concentrated burst of development. This paper reviews the recent research progress in ML-based technologies for USV autonomous decision-making and intelligent planning, and analyzes the challenges and opportunities for future research.

This paper first reviews the autonomy levels of USVs and the development history of ML in unmanned marine systems. It then …

CRediT authorship contribution statement

Zongyang Liu: Writing – original draft, Validation, Conceptualization. **Qin Zhang:** Writing – review & editing. **Xianbo Xiang:** Supervision, Project administration, Funding acquisition. **Shaolong Yang:** Investigation, Formal analysis. **Yi Huang:** Writing – original draft, Investigation. **Yanji Zhu:** Formal analysis, Conceptualization. …

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

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